OC-GS: Object-Centric Representation and Demand Alignment for Immersive 3D Gaussian Streaming

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Abstract

3D Gaussian Splatting (3DGS) enables high-fidelity immersive rendering, but scene streaming for live 6DoF viewing still depends on anticipating and prioritizing content before user attention shifts. Because the client cannot deliver the whole scene at once, the core systems problem is not simply how much data can be sent, but which content should receive the limited budget before attention moves. Current systems commonly organize requests spatially, whereas user attention often concentrates on semantic objects. This mismatch wastes transmission budget on collateral context and weakens post-saccade recovery under tight runtime constraints. In our live evaluation, one important limitation in this setting is not only prediction accuracy, but also whether the request interface can directly expose attended content.

We propose ***OC-GS*** (Object-Centric Gaussian Streaming), a two-stage framework for demand-aligned delivery. In the offline stage, semantic–saliency co-distillation converts frozen 3DGS into executable object-addressable transmission units comprising macro background nodes, hierarchical chains for large objects, and atomic micro-packs for fine details. In the online stage, object-level demand prediction and Lyapunov stability-aware control convert short-horizon demand estimates into feasible low-churn requests under coupled resource budgets. Crucially, representation, prediction, and control are coupled through the same executable object-level action space. Against native L3GS/LapisGS baselines and an A3FR variant with only a thin transport adapter, *OC-GS* primarily improves recovery and stability rather than chasing the full-resident upper bound: in live eye-tracked 6DoF Meta Quest sessions on Mip-NeRF360, ScanNet++, and Replica, it reduces recovery latency from 139–192 ms to 72–78 ms and refetch ratio from 5.4–9.4% to 0.8–2.3%, while improving Foveated-PSNR by 0.87–1.77 dB over A3FR. Our project page is available at https://anonymous.4open.science/w/OC_GS-Project-Page-0E5C/.

CCS Concepts

• **Information systems** → **Multimedia streaming**; • **Computing methodologies** → **Computer vision**; • **Human-centered computing** → **Virtual reality**.

Keywords

3D Gaussian Splatting, foveated streaming, semantic-saliency distillation, object-level demand scheduling, 6DoF volumetric media

1 Introduction

3D Gaussian Splatting (3DGS) [11] enables high-fidelity immersive rendering, but scene streaming for live 6DoF viewing must decide what to deliver before user attention shifts. Scene payloads are still substantial: in Mip-NeRF360, the *bicycle* scene alone exceeds 1.33 GB [1]. At the same time, the window for post-saccade recovery is short. The client cannot fetch the whole scene at once, nor can it keep and decode everything locally. The practical goal is therefore not to stream everything as quickly as possible, but to restore the region the user is about to attend first.

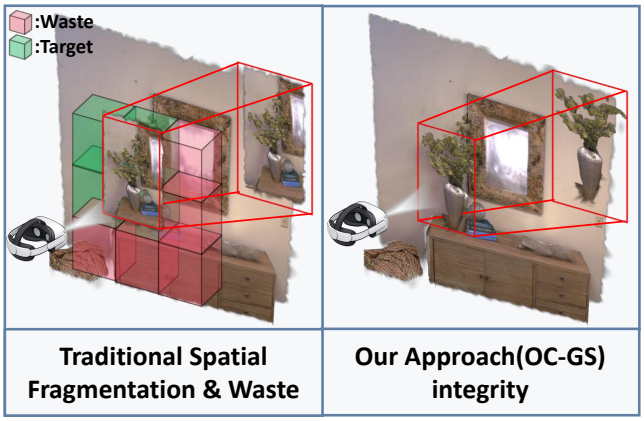

Figure 1: Representation–demand alignment. Even with approximately correct target identification, spatial-first streaming couples attended objects with surrounding context and incurs unnecessary transfer. *OC-GS* decouples them through object-addressable units derived from dual-channel distillation.

Existing systems approach this problem through spatial organization, including cuboid partitions [42], hierarchical LoD [29, 32], and tiled grids [9], defining the *request unit* as a region of 3D space. This is a natural starting point that aligns with octree structures and provides broad coverage, yet it does not directly answer a core runtime question: *under fixed budgets, what should be fetched first for the next moment of user attention?*

Our eye-tracking analysis shows that this spatial assumption often fails in live 6DoF Meta Quest viewing (Section 2). The top-3 objects capture 39.1% of foveal attention per frame, more than double the 18.75% expected under uniform allocation across ∼16 visible instances. When attention is this concentrated, spatial blocks enclosing the target inevitably bundle surrounding context, wasting bytes on collateral transfer. Under revisits, small prediction perturbations further trigger eviction–refetch oscillations that convert cache reuse into destabilizing churn. The fundamental problem is not only *where* the user may look next, but whether the transport interface can expose attended content itself as a legal, budget-feasible request target.

This analysis points to three coupled requirements that spatial-first designs struggle to satisfy simultaneously: **independent object-level addressability**, so that fetching a target does not force collateral transfer; **object-granularity demand prediction**, because spatial-tile forecasting is an indirect proxy for which object the user will attend; and **stable control under coupled budgets**,

because greedy optimization, even with accurate predictions, can still produce fetch–evict oscillations that degrade long-horizon quality.

We propose ***OC-GS*** (Object-Centric Gaussian Streaming), a two-stage framework that compiles object structure into the transport interface itself, so that object-level demand defines the executable action space in which prediction and control jointly operate. In the offline stage, semantic–saliency co-distillation over frozen 3DGS (supervised by SAM3 [2]/CLIP [24] and VST++ [15]) constructs an object-addressable hybrid-granularity substrate comprising macro background nodes, hierarchical chains for large objects, and atomic micro-packs for fine details, together with dependency DAGs, cost metadata, and refinement frontiers that define the set of legally requestable actions. In the online stage, an object-anchored predictor estimates near-term demand from head-eye history, and a Lyapunov-based scheduler converts those estimates into budget-feasible, low-churn fetch–evict decisions. Because both stages consume the same exported object-level action space, predicted demand translates directly into legal requests without intermediate spatial remapping.

We evaluate *OC-GS* on Mip-NeRF360 [1], ScanNet++ [40], and Replica [30] in live eye-tracked 6DoF sessions. L3GS [32] and LapisGS [29] run natively; A3FR [36] receives only a thin transport adapter. Our contributions are:

- **Representation–demand mismatch.** We show with controlled eye tracking that, in live 6DoF viewing, attention is usually concentrated on a small set of semantic objects. This makes spatial request units inefficient because they bring in surrounding context and amplify revisit-driven churn (Section 2).
- **Executable object-centric streaming.** We design an offline packaging scheme that exposes object-addressable transmission units for frozen 3DGS, together with explicit dependency and frontier metadata. As a result, the predictor and the scheduler operate on the same object units instead of predicting objects and then mapping them back to spatial blocks (Sections 4–5).
- **Recovery and long-horizon stability.** Against native baselines, *OC-GS* reduces recovery latency from 139–192 ms to 72–78 ms and refetch from 5.4–9.4% to 0.8–2.3%, while improving Foveated-PSNR by 0.87–1.77 dB over A3FR. Ablations further show that the full closed loop mainly improves long-horizon stability rather than the lowest single-event recovery time (Section 6).

2 Motivation: Object-Centric Demand

Beyond deciding *what* to prefetch, a 3DGS scene-streaming system must decide at *what granularity* demand is represented and acted upon. Existing systems apply FoV culling and reactive transfer yet still miss short post-saccade recovery windows, suggesting a problem that visibility filtering alone cannot resolve. We trace a four-step causal chain covering recovery gap, attention concentration, granularity overhead, and revisit churn, summarized in Fig. 2(a)–(d), and ask whether the recurring recovery failure is better explained by bandwidth deficit or by interface mismatch under tight budgets.

2.1 Post-Saccade Recovery Gap

Users judge rendering quality within a narrow post-saccade window. Fig. 2(a) shows that L3GS, LapisGS, A3FR, and heuristic reactive baselines frequently exceed the 100 ms gaze-contingent responsiveness reference established in foveated VR studies [23, 27], even under aggressive transfer. The recurring failure mode is not that these systems never raise quality, but that they restore the *wrong* content first: coarse spatial context arrives before the semantically critical details that determine perceived recovery. This pattern points to an interface-level problem in which spatial units delay the discriminative parts of attended objects even when approximate region prediction is available.

Trace-level latency decomposition is consistent with this diagnosis. A reactive cycle typically serializes attention-state update, candidate re-ranking, transfer, decode, and visible commit; even when each stage is individually brief, their composition can consume most of the post-saccade comfort window. Under spatial units, early transferred bytes are often spent on contextual regions that do not immediately increase attended detail quality, so the first perceptually meaningful gain arrives later than the first network response. Faster transport alone therefore cannot fully eliminate recovery lag when the request unit remains misaligned with attended semantics.

2.2 Object-Centric Attention Statistics

A well-established finding in visual cognition is that attentional facilitation is organized by object structure rather than isolated spatial coordinates [4, 5, 33]. The Egly–Driver–Rafal double-rectangle paradigm demonstrates a robust same-object advantage even when candidate targets are equidistant in space [5, 33], and trans-saccadic studies further show that observers preferentially preserve object continuity across gaze shifts [10, 31]. Complementary oculomotor work reveals that microsaccades and drift persist during nominal fixation [16, 17, 19], making instantaneous gaze coordinates a noisy proxy for the stable semantic target.

We verify this pattern in immersive 3DGS viewing with a cross-site 6DoF eye-tracking cohort on Mip-NeRF360 scenes (16.8 visible instances per frame on average). For each frame t, we place a Gaussian foveal kernel $G_t(p)$ at the gaze center, derive object masks $M_{t,n}(p)$ via SAM3 with temporal matching, and compute each object's share of gaze-centered mass:

$$a_{t,n} = \frac{\sum_p G_t(p) M_{t,n}(p)}{\sum_p G_t(p)}, \quad n = 1, \ldots, N_t, \tag{1}$$

where N_t is the number of visible objects. The top-K cumulative concentration is

$$C_t(K) = \sum_{n \in \mathrm{TopK}(a_t)} a_{t,n}. \tag{2}$$

After synchronizing head pose and binocular gaze and applying area-normalized, center-bias-corrected, and saliency-only controls, concentration remains pronounced: Top-1 reaches 13.0% versus a 6.25% uniform reference (corresponding to ∼16 objects), Top-3 reaches 39.1% versus 18.75%, and 67% of frames have Top-3 above 30%. Cross-participant consistency is strong (ICC = 0.84, 95% CI [0.79, 0.89]; binocular $r = 0.91$, $p < 0.001$). The implication for streaming is that a small object set carries a disproportionate share of

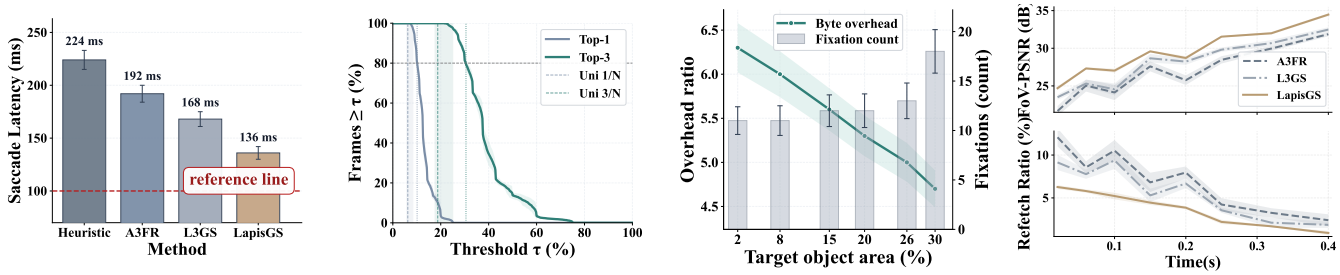

Figure 2: Motivation chain: recovery lag, object-centric attention concentration, spatial granularity overhead under fixed budgets, and revisit-driven oscillation. Panel (a) uses a 100 ms post-saccade reference line as a stringent gaze-contingent responsiveness reference [23, 27]; panel (b) reports Top-1/Top-3 object-mass concentration rather than a pixel-saliency CDF.

demand and should dominate the next request decision. Supplementary material reports control settings and mask-consistency diagnostics.

2.3 Granularity Cost of Spatial Units

Given object-centric attention, the cost of spatial request units becomes apparent. Spatial blocks cut across object boundaries, so recovering a small attended target can require transferring a much larger enclosing neighborhood (Fig. 2(c)). This is a structural *transfer-indivisibility* problem, not a prediction error: even when the target is correctly identified, the enclosing block still consumes unnecessary bandwidth. The scheduler effectively pays twice, as the current step wastes bytes on surrounding context and subsequent steps inherit less headroom for the next attended object. Improving prediction accuracy alone cannot remove this overhead as long as the transport unit continues to bundle the target with unrelated context.

The key inefficiency is an object-scale asymmetry: as attended targets become smaller or more fragmented, the ratio of useful to transferred bytes drops rapidly for fixed spatial blocks, reducing effective budget utilization precisely in the regimes where rapid detail restoration matters most. Consequently, a policy that appears feasible under gross-byte accounting can still be attention-inefficient under object-level accounting, especially in cluttered scenes with multiple small, high-saliency candidates competing within the same bandwidth epoch.

2.4 Runtime Thrashing Under Revisits

Immersive viewing frequently revisits a small set of high-value objects. Such revisits should, in principle, benefit from cache reuse. Under spatial fragmentation, however, small prediction perturbations and boundary jitter repeatedly flip block priorities, triggering eviction–refetch loops that turn beneficial reuse into policy oscillation. We quantify this with a normalized churn score:

$$\eta_t = \frac{R_t}{E_t + \epsilon}, \tag{3}$$

where R_t and E_t are refetch and eviction counts. Fig. 2(d) reveals refetch spikes and FoV-quality oscillation under spatial-first policies, reflecting churn that exceeds what demand volatility alone would explain and is instead amplified by request units that do not align with the semantic target being revisited.

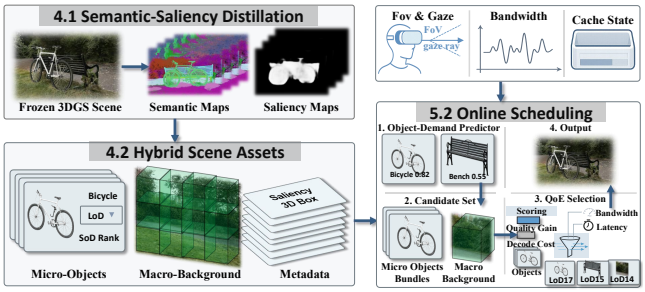

Figure 3: Two-stage *OC-GS* pipeline over a shared decision space: offline objectization exports executable units and dependencies, and online prediction/scheduling operates directly on the same object-level action space.

Once such churn appears, its effect is not purely instantaneous but cumulative. Re-fetched blocks compete with genuinely new demand, queue pressure rises, and short-horizon corrections can alternately overshoot and compensate across epochs. The resulting hysteresis manifests as repeated sharpness variation around frequently revisited objects and delayed reconvergence after attention shifts, which is perceptually more disruptive than a single isolated quality dip.

Bandwidth naturally remains a factor, but the evidence above indicates that, under limited budgets, unit design and control stability are equally important. Once the request interface fragments attended objects, stronger prediction alone cannot eliminate persistent churn. This motivates building object-addressable units offline and applying stability-aware control online, the design developed in Sections 4–5.

3 Method Overview

OC-GS is built around a central design choice: rather than relying solely on a stronger predictor, it establishes a shared executable decision space that makes object demand directly actionable under runtime legality and budget constraints. This design follows the three requirements identified in Section 2: request units must align with attended objects, demand prediction must operate at the object level, and control must remain stable under coupled runtime budgets.

Offline, frozen primitives are reorganized into macro background nodes, hierarchical object chains, and atomic micro-packs. The exported interface consists of an addressable unit list $\mathcal{U}$, dependency DAG $\mathcal{D}$, per-unit metadata $\mathcal{M}eta$, and executable frontier $\mathcal{F}_t$, so runtime decisions act on feasible requests rather than spatial proxies. In practical terms, $\mathcal{U}$ specifies what can be requested, $\mathcal{D}$ determines which requests are legal given refinement dependencies, $\mathcal{M}eta$ quantifies how each unit contributes to quality and cost, and $\mathcal{F}_t$ identifies which refinements are currently open.

Online, an object-anchored predictor estimates near-term demand from motion, geometry, and interaction history, and a Lyapunov scheduler converts those estimates into budget-feasible low-churn actions. Because predictor and scheduler consume the same exported interface, object demand does not need to be translated back into spatial guesses before transmission. Representation, prediction, and control therefore share one executable decision space.

This shared interface is the principal systems contribution of *OC-GS*. If representation remained spatial, prediction could only rank proxies rather than the attended object itself; if control ignored dependencies and refinement frontiers, object demand could not be converted into legal requests without ad hoc translation. By coupling the two stages around one decision space, *OC-GS* integrates semantic structure, prediction, and scheduling into a single closed executable loop.

4 Offline Stage: Object-Addressable Substrate

4.1 Semantic–Saliency Co-Distillation

This first offline step addresses a specific systems problem: how to attach object-assignable side signals to frozen 3DGS so that subsequent online decisions can operate on semantically coherent units rather than spatial surrogates. Fig. 4 summarizes the co-distillation pipeline. Semantics characterizes structural coherence, whereas saliency identifies what is worth exposing as a requestable unit.

For each primitive g_i, we optimize a semantic feature $\mathbf{f}_i$ and saliency value $s_i \in [0,1]$ while freezing base 3DGS parameters. To avoid dense per-primitive semantics, we reconstruct features from a shared codebook S and sparse coefficients $\boldsymbol{\beta}_i$:

$$\mathbf{f}_i = \boldsymbol{\beta}_i^\top S, \tag{4}$$

where $S \in \mathbb{R}^{L\times D}$ is a shared codebook with $L = 256$ and $D = 64$; $\mathbf{f}_i$ is normalized to $\hat{\mathbf{f}}_i$ for angular supervision.

We supervise both channels through the same alpha-compositing visibility weights used by the base 3DGS rasterizer. For pixel p in view v, the rendered semantic direction is

$$\mathbf{R}_v(p) = \frac{\sum_{i=1}^{K_p} \omega_i^v(p)\hat{\mathbf{f}}_{g_i^p}}{\sum_{i=1}^{K_p} \omega_i^v(p) + \varepsilon}, \tag{5}$$

where $\omega_i^v(p)$ is the visibility weight induced by fixed alpha compositing. The same projection is applied to saliency.

The semantic branch uses **SAM3** [2] masks and **CLIP-ViT-L/14** [24] targets for in-region consistency and inter-instance separation, while the saliency branch aligns rendered saliency to the **VST++** teacher [15]:

$$\hat{A}_v(p) = \frac{\sum_{i=1}^{K_p} \omega_i^v(p) s_{g_i^p}}{\sum_{i=1}^{K_p} \omega_i^v(p) + \varepsilon}. \tag{6}$$

A shared rendering path keeps the two channels geometrically consistent.

We train these side attributes with a joint objective for semantic discrimination, saliency alignment, and sparse coding:

$$L_{\text{off}} = \lambda_{\text{sem}} L_{\text{sem}} + \lambda_{\text{sod}} L_{\text{sod}} + \lambda_{\text{sp}} \sum_i \|\boldsymbol{\beta}_i\|_1 + \lambda_{\text{ent}} L_{\text{ent}}. \tag{7}$$

Here, L_{sem} supervises semantic logits, L_{sod} aligns $\hat{A}_v$ with the teacher saliency map, and the remaining terms enforce sparse, non-collapsed coding. We update only S, $\{\boldsymbol{\beta}_i\}$, and $\{s_i\}$; base geometry and appearance remain frozen.

The codebook parameterization reduces payload size to approximately 12.4 MB plus a 64 KB shared table at $N{=}10^6$, lowering metadata transfer and decode overhead. This figure refers to codebook-based side representation accounting before final packed export and indicates the compactness scale of the side channel.

After convergence, we harden top-K assignments and quantize retained coefficients. The exported payload contains the shared codebook, sparse per-Gaussian indices/weights, and saliency scalars. The practical transmitted semantic-side payload used in all experiments is this final packed export (Table 1), i.e., 6.2 MB + 64 KB.

4.2 Hybrid Decoupling and Classification

This stage addresses the next offline problem: converting object-worthiness into runtime-consumable units with explicit queue semantics, rather than leaving the controller to reason over raw primitives. We convert distilled side signals into explicit object structure, beginning with a local reliability score that measures semantic agreement among neighboring primitives.

$$r_i = |\mathcal{N}(i)|^{-1} \sum_{j\in\mathcal{N}(i)} \max(0, \hat{\mathbf{f}}_i^\top \hat{\mathbf{f}}_j), \tag{8}$$

where $\mathcal{N}(i)$ is the 3D neighborhood of g_i (radius 0.1m).

We then cluster reliable primitives using a distance that combines spatial proximity and semantic dissimilarity:

$$d(i,j) = \rho^{-2}\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2^2 + \lambda\big(1 - \hat{\mathbf{f}}_i^\top \hat{\mathbf{f}}_j\big). \tag{9}$$

Once clusters are formed, we route them into three queues: Queue A for macro background, Queue B for large objects with hierarchy, and Queue C for compact atomic packs. Queue A provides stable coarse coverage, Queue B supports progressive refinement for large objects, and Queue C enables direct atomic access for fine details. Object-worthiness is evaluated as

$$w_m = \bar{s}_m \kappa_m, \tag{10}$$

where $\bar{s}_m$ is mean saliency and κ_m is semantic reliability. Clusters with low w_m remain in Queue A, while retained clusters are assigned according to their scene-normalized scale $h_m = \max(\tilde{v}_m, \tilde{b}_m)$: large or heavy objects are routed to Queue B and compact ones to Queue C. A single validation-selected operating point is fixed across all scenes and sites; supplementary material reports sensitivity sweeps. For each retained cluster, we derive saliency-weighted descriptors:

$$\bar{\mathbf{f}}_m = \frac{\sum_{i\in O_m} s_i \hat{\mathbf{f}}_i}{\left\|\sum_{i\in O_m} s_i \hat{\mathbf{f}}_i\right\|_2 + \varepsilon}, \qquad \bar{s}_m = |O_m|^{-1} \sum_{i\in O_m} s_i. \tag{11}$$

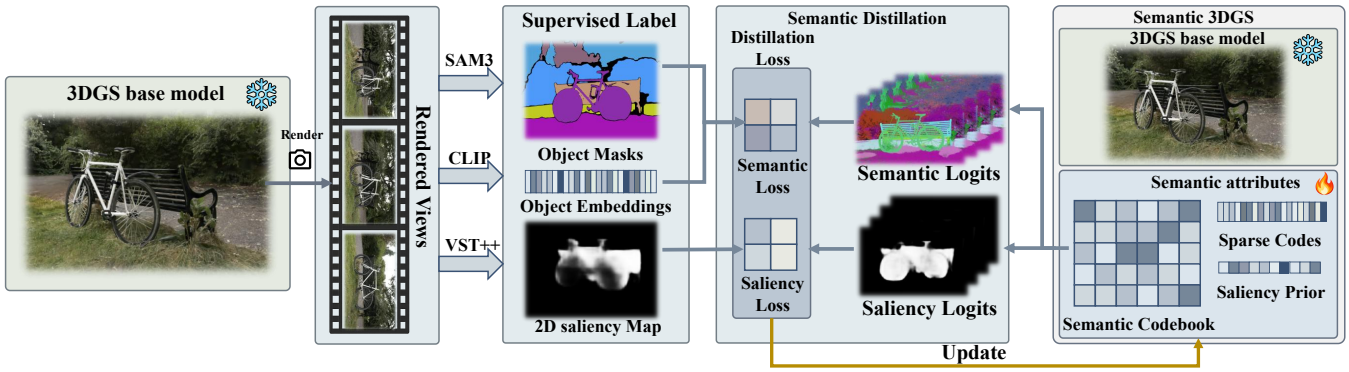

Figure 4: Offline co-distillation on frozen 3DGS: semantic and saliency teachers supervise rendered logits, while only side parameters are updated.

Eq. (11) yields compact object metadata directly consumed by the online predictor and scheduler.

4.3 Runtime Packaging

This final offline step resolves granularity conflicts prior to runtime. After objectization, Queue A is packaged as macro octree LoD nodes for stable coverage, Queue B as coarse-to-fine intra-object chains, and Queue C as atomic micro-object assets. Packaging therefore governs not only storage organization but also the granularity and legality of all subsequent online actions, enabling the runtime loop to focus on legal selection under dynamic budgets rather than on late structural repair.

5 Online Stage: Stability-Driven Control

5.1 Object-Anchored Demand Prediction

At time t, the predictor outputs the object-demand distribution $p_{\text{total}}(o \mid t)$ over the object-addressable entities exported offline (Queues B/C), while Queue-A macro candidates are derived from geometry-based coverage and merged into the same loop. Its role is not to define a separate contribution axis, but to consume the same exported object interface as the scheduler, so that demand no longer requires translation back into spatial guesses before control.

The predictor consumes synchronized head-eye history, object descriptors, and runtime state. Three complementary evidence streams operate over the same object set: a physical branch (2-layer Transformer over 20 frames) extrapolates motion; a visual branch scores geometric imminence using frustum overlap, projected coverage, ray distance, and occlusion cues; and an intent branch summarizes recent hit-object sequences after run-length compression, 80 ms dwell filtering, and background pruning. These streams are fused to estimate object-level demand on the exported interface:

$$p_{\text{total}}(o \mid t) = \text{softmax}\big(\alpha_t \log(p_{\text{geom}}(o \mid t) + \varepsilon) + (1-\alpha_t)\log(p_{\text{intent}}(o \mid t) + \varepsilon)\big), \tag{12}$$

where $\alpha_t = \sigma(\text{motion_var})$, so that stable motion favors geometry and uncertain motion favors intent: under smooth motion, future visibility is largely geometry-determined, whereas under abrupt gaze or head changes, recent interaction history becomes more informative. The branch split serves to estimate object demand in the same decision space subsequently used by the controller. Training uses $L_{\text{on}} = \lambda_{\text{kin}} L_{\text{kin}} + \lambda_{\text{cls}} L_{\text{cls}} + \lambda_{\text{unc}} L_{\text{unc}}$, with prediction horizon $\Delta_p = 100$ ms and scheduler epoch $\Delta_s = 100$ ms.

5.2 Lyapunov QoE Scheduling

Every Δ_s seconds, the scheduler examines the object frontier $\mathcal{F}_t$ and macro candidates $\mathcal{K}_{\text{mac}}(t)$, and optimizes over their union. The binary decision vector x_t is the per-epoch request action: $x_k(t) = 1$ means unit k is requested in the current epoch, and $x_k(t) = 0$ means it is skipped. This control layer is necessary because short-horizon demand gains alone can still drive aggressive fetch/evict cycles under revisits; the scheduler must convert immediate demand into sustainable, low-churn request behavior. This is precisely the distinction isolated in Table 3: a stronger predictor can lower per-event latency while still leaving refetch and temporal instability high.

$$x_k(t) = 1 \Rightarrow \text{pred}_{\mathcal{D}}(k) \subseteq \mathcal{M}(t), \quad \forall k. \tag{13}$$

Here $\mathcal{M}(t)$ is the resident set at the start of epoch t, and $\text{pred}_{\mathcal{D}}(k)$ is the set of prerequisite units of k in the offline dependency DAG. Eq. (13) states that the scheduler may request only currently executable frontier units, not arbitrary descendants.

For each feasible action, we score attended benefit against client-side cost:

$$\text{QoE}(t) = f_t(x_t) - \lambda g_t(x_t), \tag{14}$$

where f_t is expected attended-quality gain and g_t is client processing load:

$$f_t(x_t) = Q^{\text{base}}(t) + \sum_{k \in \mathcal{F}_t} x_k p_{\text{total}}(o(k) \mid t) \Delta Q_k(t) + \sum_{k \in \mathcal{K}_{\text{mac}}(t)} x_k v_k(t) \Delta Q_k(t), \tag{15}$$

$$g_t(x_t) = \sum_k x_k \, \text{dec}_k. \tag{16}$$

Here $Q^{\text{base}}(t)$ is the attended quality if no new unit is admitted in the current epoch. For an object-frontier unit $k \in \mathcal{F}_t$, $o(k)$ denotes its parent object, $p_{\text{total}}(o(k) \mid t)$ is the predicted demand probability of that object, and $\Delta Q_k(t)$ is the marginal quality gain if k is fetched in the current epoch. For a Queue-A macro unit $k \in \mathcal{K}_{\text{mac}}(t)$, $v_k(t)$

is a geometry-based coverage weight. Eq. (16) sums the per-unit decode cost dec_k across all selected units.

The action must satisfy instantaneous bandwidth and decoding budgets:

$$\sum_k \mathrm{size}_k x_k(t) \le B(t)\Delta_s, \tag{17}$$

$$\sum_k \mathrm{dec}_k x_k(t) \le C_{\max}(t). \tag{18}$$

Here size_k is the transfer size in bytes, $B(t)\Delta_s$ is the bandwidth budget available during the current epoch, and $C_{\max}(t)$ is the permitted client decode budget. To suppress bursty cross-epoch behavior, we maintain virtual queues $Z_B(t)$ and $Z_C(t)$ that summarize recent bandwidth and decode pressure, with Lyapunov state $\Theta_t = (Z_B(t), Z_C(t))$, function $L(\Theta_t) = \frac{1}{2}(Z_B^2(t) + Z_C^2(t))$, and one-step drift $\Delta(\Theta_t)$:

$$Z_B(t{+}1) = \Big[Z_B(t) + \sum_k \mathrm{size}_k x_k(t) - B(t)\Delta_s\Big]^+, \tag{19}$$

$$Z_C(t{+}1) = \Big[Z_C(t) + \sum_k \mathrm{dec}_k x_k(t) - C_{\max}(t)\Big]^+. \tag{20}$$

Here $[\cdot]^+ = \max(\cdot, 0)$. These are virtual control states rather than physical packet queues: they grow when recent actions have been too aggressive relative to bandwidth or decode capacity, and shrink when the scheduler returns to a sustainable operating point. Applying the standard one-step drift upper bound, long-horizon control reduces to the following per-slot drift-plus-penalty objective:

$$x_t^\star \in \arg\max_x J_t(x), \tag{21}$$

$$J_t(x) = V\big(f_t(x) - \lambda g_t(x)\big) - Z_B(t)\sum_k \mathrm{size}_k x_k - Z_C(t)\sum_k \mathrm{dec}_k x_k. \tag{22}$$

The maximization is subject to Eq. (13) and Eqs. (17)–(18). Eq. (22) makes the trade-off explicit: the first term rewards immediate attended-quality gain, while the Z_B and Z_C terms penalize actions that persistently oversubscribe bandwidth or decode capacity across epochs. Lyapunov control is adopted here not to claim global optimality, but to prevent short-term quality optimization from destabilizing long-horizon streaming behavior under revisits and coupled budgets. We set $V(t) = 1 + \alpha \max(Z_B(t), Z_C(t))$ with $\alpha = 0.01$, so that larger virtual queues directly cause the controller to become more conservative.

In practice, we solve Eq. (21) via rank-and-scan: each candidate unit's contribution to $J_t(x)$ is computed and sorted once, and units are greedily added while preserving dependency and budget feasibility. We adopt this strategy because the feasible frontier per epoch is small and the objective is sufficiently monotonic for efficient greedy approximation under runtime constraints.

If cache overflow occurs after selecting x_t, we evict the lowest-retention non-pinned resident blocks according to $\phi_k(t) = p_{\mathrm{total}}(o(k) \mid t)\Delta Q_k(t) - \gamma_a \mathrm{age}_k - \gamma_r \mathbb{I}[k \in \mathcal{R}_{\mathrm{recent}}]$. Dependency ancestors and newly fetched blocks are pinned for one epoch, and the remaining non-resident selected blocks form $\mathcal{Q}(t)$. In all main experiments, we use a 3-epoch recent window and 1-epoch pinning; supplementary material reports sensitivity sweeps over queue weights and pinning policy.

6 Evaluation

6.1 Implementation

We implement *OC-GS* in PyTorch with a differentiable Gaussian rasterizer [11]. Offline distillation and packaging run on an Intel Core i9-13900K with an NVIDIA RTX 3090 (24 GB), requiring approximately 3 hours per scene; the exported substrate is scene-specific and reused across users. Online, participants use a Meta Quest headset with no per-user retraining; the predictor (2-layer Transformer, 4 heads, hidden size 128) runs in under 10 ms, and the scheduler runs every 100 ms with rank-and-feasibility greedy selection. Table 1 summarizes the practical overhead of *OC-GS*. The one-time offline objectization cost is amortized across users, while online control remains lightweight relative to the 100 ms scheduling epoch. For fairness, we synchronize clocks, inject the same bandwidth traces, and log the same event types across all methods; L3GS and LapisGS retain their native hierarchy and controller stacks, while A3FR receives only a thin transport adapter.

Table 1: Practical overhead summary of *OC-GS*. One-time offline costs are amortized across users, while online control overhead remains lightweight relative to the 100 ms scheduling epoch.

Item	Cost
Offline distillation + packaging (per scene)	~3 hours
Online predictor latency (per step)	$<$ 10 ms
Online scheduler period	100 ms
Semantic side payload	6.2 MB + 64 KB
Dense semantic equivalent (N=10^6)	~244 MB
Added shared cold start @ 100 Mbps	0.6 s
Added shared cold start @ 20 Mbps	2.7 s

6.2 Baselines and Metrics

Our comparison is native-first rather than wrapper-normalized. We compare *OC-GS* against the original 3DGS (full-resident visual-quality upper bound) and three reproduced baselines: L3GS, LapisGS, and A3FR. L3GS and LapisGS are run in native form with their original hierarchy and scoring logic. Because A3FR is originally a gaze-conditioned rendering system rather than a network-delivery stack, we attach only a thin transport adapter that forwards network-delivered content while preserving its gaze-conditioned prioritization.

The study spans two sites and Meta Quest participants on scenes drawn from Mip-NeRF360 [1], ScanNet++ [40], and Replica [30]. Predictor training uses user-disjoint and scene-disjoint splits, with held-out users and scenes reserved for testing. Figures and tables report results on the held-out subset with hyperparameters fixed before test-time evaluation.

Primary metrics are Foveated-PSNR, SSIM, recovery latency, and refetch ratio (Table 2); Fig. 5 additionally reports 3D fixation-point final displacement error (FDE), and Table 3 reports Temporal Instability.

6.3 Quantitative Results

The primary benefit of *OC-GS* is not closing the gap to the full-resident upper bound, but converting a limited budget into faster

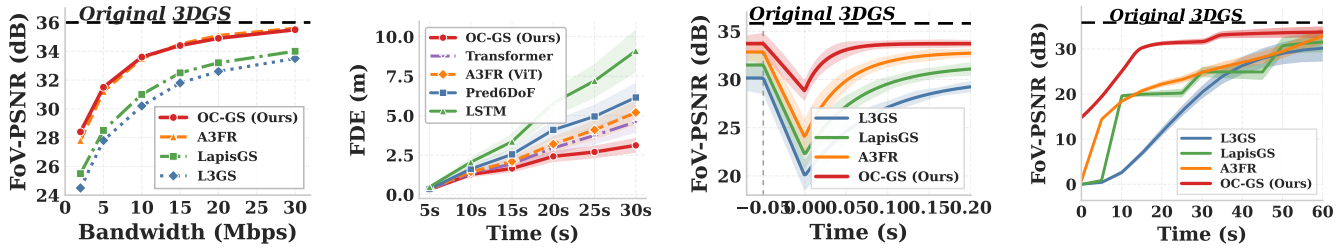

Figure 5: Main results on rate-distortion, predictive accuracy, post-saccade recovery, and long-session stability. In all figures and tables, *Original 3DGS* denotes the theoretical upper bound under fully transmitted/full-resident content.

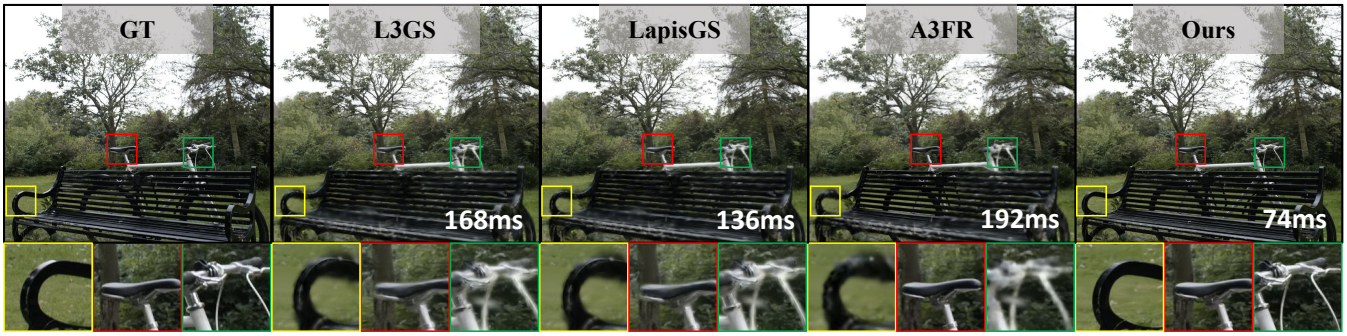

Figure 6: Qualitative comparison under the same event and the same budget at a fixed 50 ms after gaze/saccade shift.

Table 2: Main comparison on Mip-NeRF360 [1], ScanNet++ [40], and Replica [30] using Fov-PSNR, SSIM, Recovery latency, and Refetch ratio (dataset averages; best/second highlighted).

Method	Mip-NeRF360 [1]				ScanNet++ [40]				Replica [30]			
	Fov-PSNR ↑	SSIM ↑	Recovery (ms) ↓	Refetch ↓	Fov-PSNR ↑	SSIM ↑	Recovery (ms) ↓	Refetch ↓	Fov-PSNR ↑	SSIM ↑	Recovery (ms) ↓	Refetch ↓
Original 3DGS	35.82	0.982	N/A	N/A	38.44	0.986	N/A	N/A	37.18	0.984	N/A	N/A
L3GS [32]	30.15	0.895	168	8.4%	32.68	0.921	161	12.7%	31.27	0.907	171	13.5%
LapisGS [29]	31.50	0.910	136	4.2%	34.23	0.961	138	7.3%	32.91	0.937	147	7.9%
A3FR [36]	32.85	0.925	192	5.4%	35.68	0.972	184	8.8%	34.18	0.946	139	9.4%
OC-GS	**33.72**	**0.948**	**74**	**0.8%**	**37.45**	**0.981**	**72**	**1.9%**	**35.49**	**0.963**	**78**	**2.3%**

and more stable attended recovery. On the held-out evaluation, *OC-GS* ranks highest among all streaming baselines on all four main metrics in Table 2. Relative to A3FR, *OC-GS* reduces recovery latency from 139–192 ms to 72–78 ms and refetch from 5.4–9.4% to 0.8–2.3%, while still improving Foveated-PSNR by 0.87–1.77 dB and SSIM by 0.009–0.023. Fig. 5 shows that the same advantage persists across bandwidth sweeps and long-session trajectories. These gains are consistent across Mip-NeRF360, ScanNet++, and Replica, with the largest relative margins appearing in recovery and refetch under tighter budgets.

This behavior is consistent with the interface-level claim. Even when native baselines retain their own hierarchy and controller logic, stronger prediction alone can reduce event-wise delay, but spatial request units still induce collateral transfer and revisit churn when attention repeatedly returns to small semantic targets. By contrast, *OC-GS* combines object-addressable actions with stability-aware control, so each predicted demand increment is translated into legally feasible, low-churn updates. Consequently, the largest gains appear in recovery latency, refetch suppression, and stability.

Table 3 sharpens this point beyond the main averages. Row (d) Full-pred achieves the lowest event-wise recovery time (34.2 ms), yet still incurs 5.1% refetch and 0.96 Temporal Instability, whereas row (g) *OC-GS* reduces refetch to 0.8% and instability to 0.51. This contrast indicates that stronger prediction primarily optimizes single-event recovery, while the full *OC-GS* loop stabilizes long-horizon control under revisits and coupled budgets.

6.4 Qualitative Results

Fig. 6 provides an event-aligned snapshot at a fixed 50 ms after a gaze or saccade shift under the same transfer budget. This setting emphasizes early-recovery behavior rather than eventual convergence and complements the average metrics in Table 2. The visual trend aligns with the quantitative result that object-level actions mainly improve recovery quality in the first response window.

Compared with L3GS, LapisGS, and the thin-adapter A3FR variant, *OC-GS* restores attended object regions with cleaner boundaries and fewer residual artifacts at the same wall-clock time, especially on small high-saliency structures where collateral transfer and revisit-induced switching are costly.

6.5 Ablation

Table 3 is a necessity ladder rather than a standard drop-one ablation. Row (a) retains the original spatial units and reactive control; (b) changes only the request unit to object-addressable packages; (c) further adds geometry-based object anchoring; and (d) adds the full geometry+intent predictor but still uses myopic scheduling. Rows (e) and (f) are targeted controls: (e) uses segmentation and saliency priority without executable object-level frontier control, while (f) retains object lists and dependency checks but removes explicit stability coupling.

The key contrast is between (d) Full-pred and (g) *OC-GS*. Full-pred already reduces event-wise recovery to 34.2 ms, but still shows 5.1% refetch and higher temporal instability (0.96). *OC-GS* reduces refetch to 0.8% and instability to 0.51, indicating that predictor strength alone does not guarantee stable long-horizon behavior in this setting.

The progression from (a) to (g) therefore supports a systems conclusion: neither heuristic semantic ranking nor dependency legality alone reproduces stable long-horizon behavior; only the full executable loop consistently converts predicted demand into sustainable request behavior.

Table 3: System ablation and targeted controls.

Method	Fov-PSNR ↑	Temp. Instab. ↓	Recovery (ms) ↓	Refetch ↓
(a) Spatial	30.45	1.84	320	6.2%
(b) Obj-gran	32.10	1.36	285	4.8%
(c) Geom-anchor	32.74	1.18	162	4.4%
(d) Full-pred	33.50	0.96	**34.2**	5.1%
(e) Seg-Prio	32.62	1.14	128	3.3%
(f) OL+DepChk	32.79	1.06	112	2.6%
(g) *OC-GS*	**33.72**	**0.51**	74	**0.8%**

7 Related Work

7.1 3DGS Compression and Streaming

3DGS [11] and related radiance-field methods [8, 18, 20, 41] enable real-time rendering, but their explicit scene parameters remain heavy for streaming [32]. Quantization and codebook-based variants [21, 22, 34] reduce static storage, yet typically assume full-scene availability rather than a runtime request interface. Static-scene 3DGS streaming uses octrees, layered hierarchies, or progressive partitions, while related volumetric-video delivery relies on tiled transport and bitrate adaptation [9, 13, 26, 28, 32, 35, 43]. Across these designs, request units remain spatial and can fragment attended objects.

7.2 Gaze and Object Cues

Gaze-guided rendering and prediction exploit non-uniform visual sensitivity [3, 6, 7, 23, 36–38], while SAM/SAM2, LangSplat, and Gaussian Grouping provide strong object cues [12, 14, 25, 39]. These directions are typically applied to view-adaptive rendering, semantic labeling, or object grouping rather than to directly schedulable transmission units with dependency structure. Prior object- and grouping-based works provide cues, prior gaze works provide demand signals, and prior streaming systems provide transport abstractions; our contribution is to compile these ingredients into a single executable control interface. Among the evaluated baselines, A3FR is the closest gaze-conditioned method, but it remains spatial-first at the request-interface level; accordingly, we attach only a thin transport adapter rather than rewriting it into a common scheduler stack. *OC-GS* instead converts object cues into executable units, dependencies, and frontier metadata so that demand prediction and budgeted online control operate within the same decision space.

8 Conclusion

Immersive 3DGS streaming benefits from aligning representation, demand estimation, and control on a shared executable object-level interface. Under native-baseline evaluation, *OC-GS* improves Foveated-PSNR over A3FR by 0.87–1.77 dB, reduces recovery latency by 44–61%, and reduces refetch by 76–85%; Table 3 demonstrates that the full closed-loop design primarily improves long-horizon stability. Boundary cases and dynamic-scene extensions are deferred to the supplementary material.

From a systems perspective, the central result is not a single-metric gain but a better operating point under coupled constraints. The ablation ladder shows that object-addressable units, geometry-aware anchoring, and demand prediction each improve parts of the pipeline, yet stable long-horizon behavior appears only when prediction and control are coupled through executable frontier-constrained actions. This interface-level coupling reduces policy churn under revisits, improves budget utilization for semantically salient details, and keeps online control lightweight enough for practical 100 ms scheduling.

These findings suggest a broader design principle for interactive volumetric delivery: representation granularity should be selected jointly with online control objectives rather than optimized in isolation. In our setting, semantic units provide the right decision primitive because user attention is object-concentrated and revisit-heavy. Future work will extend this formulation to dynamic scenes with evolving topology, cross-user shared priors under stronger personalization constraints, and tighter integration with codec/runtime co-design so that transport, decoding, and control can be optimized in a single closed loop.

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